WORKED SOLUTION

STEP II 1992, Question 15

Probability and Statistics • Random Walks

STEP II 1992, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q15

The key idea

If the point makes R positive steps out of s, it ends at $x = R - (s - R) = 2R - s$, and R has a binomial distribution. So every probability here is a sum of binomial coefficients. Parity does a lot of the work: after an even number of steps only $x = 2$ and $x = 4$ are possible, and after an odd number only $x = 3$ and $x = 5$, so each $\mathrm{P}(s)$ is a sum of just two neighbouring binomial coefficients. Pascal's rule then collapses that sum to a single coefficient. For the maximum, look at the ratio of consecutive terms, separately for even and odd s.

Solution

Let R be the number of steps in the positive direction among the first s. The steps are independent, each positive with probability $\frac{1}{2}$, so $R \sim \mathrm{B}(s, \frac{1}{2})$ and

$$\mathrm{P}(R = r) = \binom{s}{r}\left(\frac{1}{2}\right)^s.$$

The position after s steps is $X = 2R - s$, so $X = x$ exactly when $R = \frac{1}{2}(s + x)$. In particular X has the same parity as s.

The values $\mathrm{P}(5)$ and $\mathrm{P}(6)$

With $s = 5$, only $x = 3$ and $x = 5$ can happen, needing $R = 4$ and $R = 5$:

$$\mathrm{P}(5) = \left[\binom{5}{4} + \binom{5}{5}\right]\frac{1}{2^5} = \frac{5+1}{32} = \frac{3}{16}.$$

With $s = 6$, only $x = 2$ and $x = 4$ can happen, needing $R = 4$ and $R = 5$:

$$\mathrm{P}(6) = \left[\binom{6}{4} + \binom{6}{5}\right]\frac{1}{2^6} = \frac{15+6}{64} = \frac{21}{64}.$$

The general even case

For $s = 2k$, the possible target points are $x = 2$ and $x = 4$, which need $R = k+1$ and $R = k+2$. By Pascal's rule $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$,

$$\mathrm{P}(2k) = \left[\binom{2k}{k+1} + \binom{2k}{k+2}\right]\left(\frac{1}{2}\right)^{2k} = \binom{2k+1}{k+2}\left(\frac{1}{2}\right)^{2k} = \binom{2k+1}{k-1}\left(\frac{1}{2}\right)^{2k},$$

using $\binom{n}{r} = \binom{n}{n-r}$ with $(2k+1) - (k+2) = k-1$ in the last step. This is the required result. It holds for $k = 1$ too: there $\binom{2}{3} = 0$, so the $x = 4$ term is simply absent, and the formula gives $\frac{1}{4}$, the chance of two positive steps.

The odd case

For $s = 2k+1$ the target points are $x = 3$ and $x = 5$, needing $R = k+2$ and $R = k+3$. In the same way,

$$\mathrm{P}(2k+1) = \left[\binom{2k+1}{k+2} + \binom{2k+1}{k+3}\right]\left(\frac{1}{2}\right)^{2k+1} = \binom{2k+2}{k-1}\left(\frac{1}{2}\right)^{2k+1}.$$

For $k = 2$ this gives $\binom{6}{1}/32 = \frac{3}{16}$, agreeing with $\mathrm{P}(5)$.

The greatest value

Write $a_k = \mathrm{P}(2k)$ for $k \geq 1$ and $b_k = \mathrm{P}(2k+1)$ for $k \geq 1$ (note $b_0 = \mathrm{P}(1) = 0$). All of these are positive, so we can compare consecutive terms by their ratio.

Ask yourself How do we simplify a ratio of two binomial coefficients?

Write both in factorials and cancel. For the even terms,

$$\frac{\binom{2k+3}{k}}{\binom{2k+1}{k-1}} = \frac{(2k+3)!}{k!\,(k+3)!} \cdot \frac{(k-1)!\,(k+2)!}{(2k+1)!} = \frac{(2k+3)(2k+2)}{k(k+3)},$$

so

$$\frac{a_{k+1}}{a_k} = \frac{(2k+3)(2k+2)}{4k(k+3)} = \frac{4k^2+10k+6}{4k^2+12k}.$$

Since the denominator is positive, this ratio is greater than 1 exactly when $10k+6 > 12k$, that is $k < 3$; it equals 1 when $k = 3$; and it is less than 1 when $k > 3$. Hence

$$a_1 < a_2 < a_3 = a_4 > a_5 > a_6 > \cdots,$$

and the largest even-step value is $a_3 = a_4 = \mathrm{P}(6) = \mathrm{P}(8) = \frac{21}{64}$. (Check: $\mathrm{P}(8) = \binom{9}{3}/2^8 = 84/256 = 21/64$.)

For the odd terms the same cancellation gives

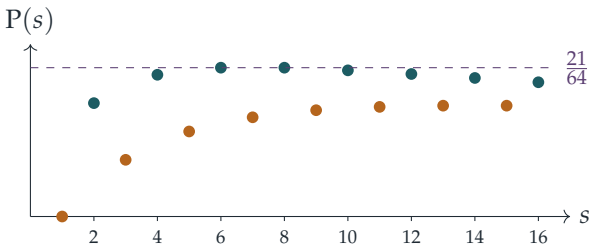
$$\frac{b_{k+1}}{b_k} = \frac{(2k+4)(2k+3)}{4k(k+4)} = \frac{4k^2+14k+12}{4k^2+16k},$$

which is greater than 1 for $k < 6$, equal to 1 at $k = 6$, and less than 1 for $k > 6$. So the largest odd-step value is

$$b_6 = b_7 = \mathrm{P}(13) = \mathrm{P}(15) = \binom{14}{5}\frac{1}{2^{13}} = \frac{2002}{8192} = \frac{1001}{4096}.$$

Finally, $\frac{21}{64} = \frac{1344}{4096} > \frac{1001}{4096}$, so no odd value beats the best even one. Therefore $\mathrm{P}(s)$ takes its greatest value, $\frac{21}{64}$, when

$$s = 6 \quad \text{and} \quad s = 8.$$



The teal points are even s and the amber points odd s: two separate sequences, each rising then falling, with the even one always on top.

Checks

- Small cases by hand: $\mathrm{P}(2) = \frac{1}{4}$ (two steps right), $\mathrm{P}(3) = \frac{1}{8}$ (three steps right), and with $k = 1$ the formulas give $\binom{3}{0}/2^2 = \frac{1}{4}$ and $\binom{4}{0}/2^3 = \frac{1}{8}$, as they should.
- An exact computer enumeration of the walk (fractions, $s \leq 60$) confirms both formulas, $\mathrm{P}(6) = \mathrm{P}(8) = \frac{21}{64} \approx 0.328$ as the overall maximum, and $\mathrm{P}(13) = \mathrm{P}(15) = \frac{1001}{4096} \approx 0.244$ as the odd maximum.

Common slips

- Adding all four target points for every s: half of them are impossible on parity grounds.
- Finding the even maximum and stopping. The odd values form a separate sequence, and you must check that its peak is lower.
- Stopping at "ratio > 1 for $k < 3$" and giving only $s = 6$: the ratio equals 1 at $k = 3$, so $s = 8$ ties.

Beyond STEP

This is the simple symmetric random walk, one of the central objects of probability. Its position after s steps spreads out like $\sqrt{s}$, so the chance of being in any fixed finite

window first rises and eventually falls to zero; the peak near $s = 7$ is where the spread is comparable to the window. The same binomial counting leads to the reflection principle and to the fact that a one-dimensional walk returns to the origin with probability 1. Grimmett and Welsh, *Probability: An Introduction*, has a good chapter on random walks.

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